

EE4730TU High Frequency Wireless Architectures

Project2: Antenna Measurements at Sub-THz Frequencies

Dr. Maria Alonso-delPino
Assistant Professor



Faculty: Electrical Engineering, Mathematics & Computer Science

Department: Microelectronics

Group: Terahertz Sensing

T +31 (0) 15 27 89307

Building 36, Room:HB18.290

m.alonsodelpino@tudelft.nl

Mekelweg 4, 2628CD Delft

<http://terahertz.tudelft.nl>

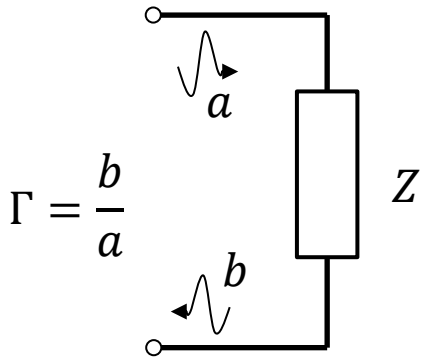
Project #2 – Antenna measurements

- Understand the HF equipment and its manipulation
- Understand the calibration procedures and limitations
- Evaluate of the coupling of two antennas in the far-field.
 - Radiation Pattern
 - Directivity
 - Gain
- Apply time-gating techniques to reduce the multiple reflections of the environment

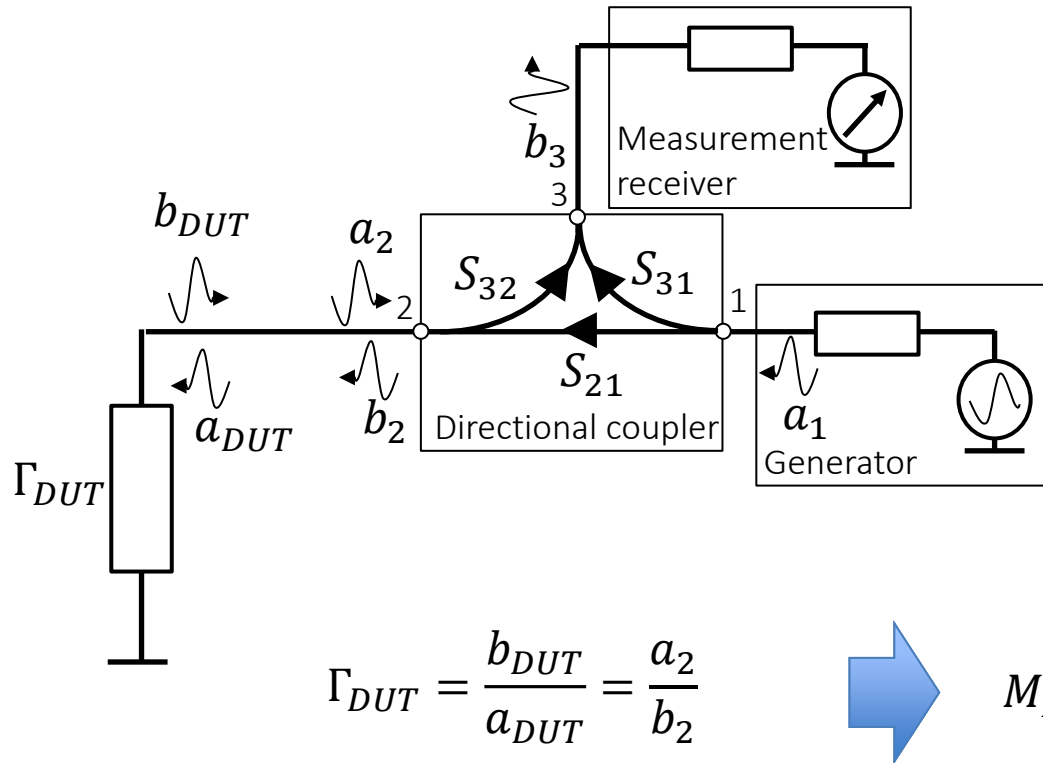


HF Instrumentation: VNA

- The most complex and versatile piece of test equipment in the field of RF engineering.
- It can measure incident waves a and reflected waves b .
- Full system error correction. Compensate the systematic measurement errors with the greatest possible precision

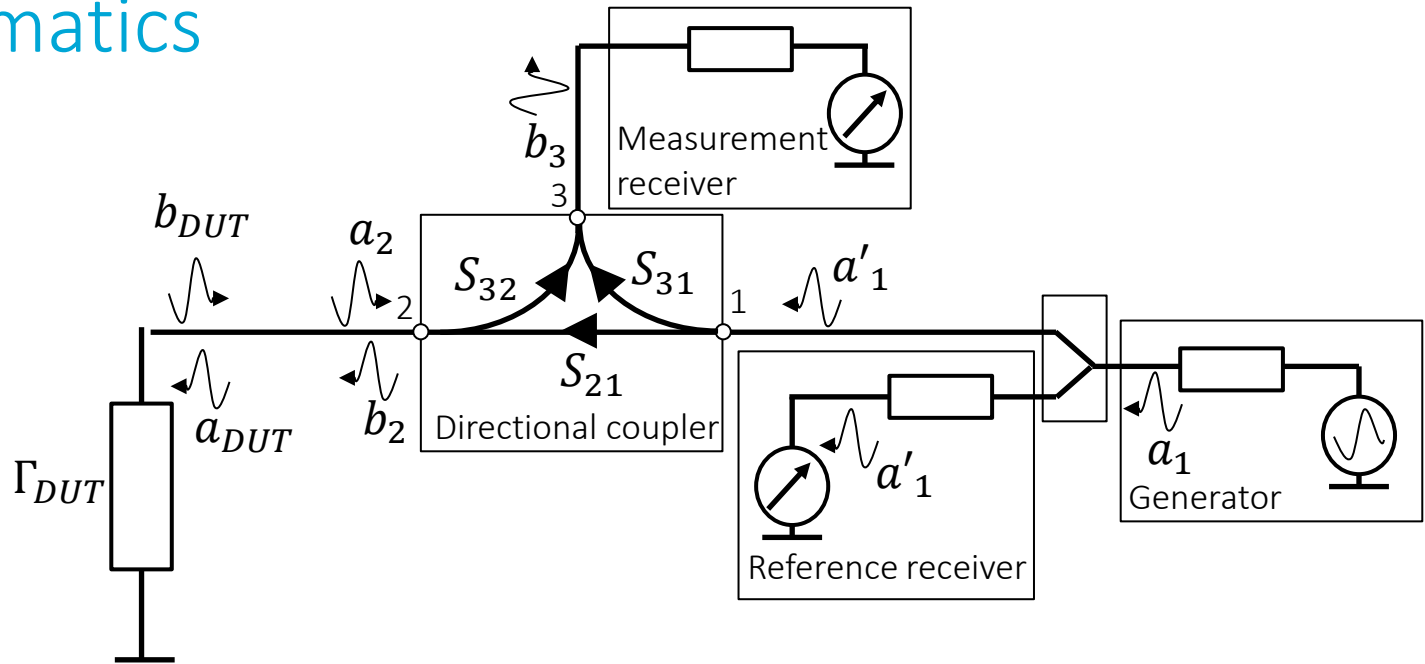


VNA – Measuring a and b waves



Assumes that the generator is maintained at a constant level

VNA - Schematics



What we measure:

$$M = \frac{b_3}{a'_1}$$

- Calibrate loss within the system:

$$L = S_{32}S_{21} \quad \Rightarrow \quad M = L \Gamma_{DUT}$$

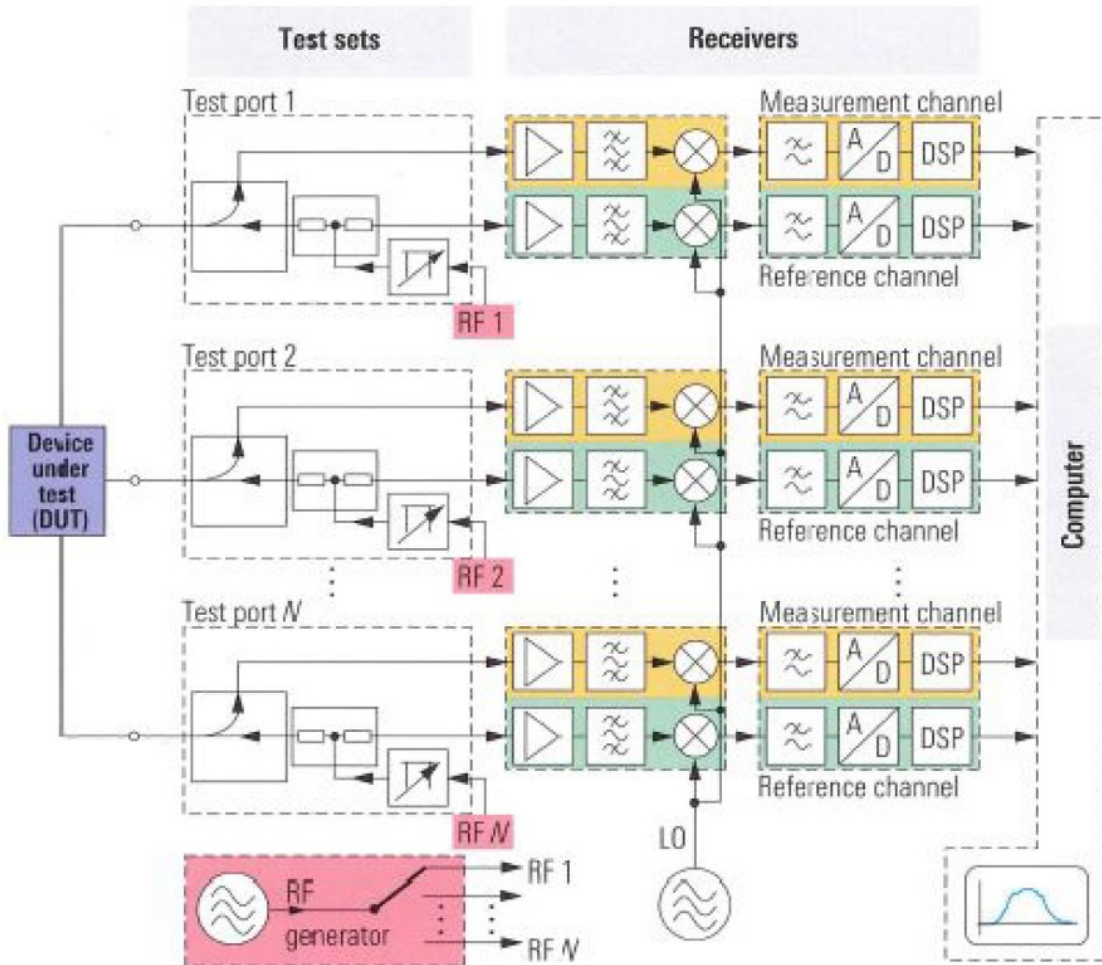
- Adjustment for the directivity of the directional element:

$$D = S_{31}/L \quad M = L(\Gamma_{DUT} + D)$$

- Adjustment for the test port reflection coefficient (S):

$$M = L(D + \Gamma_{DUT}(1 + S \times \Gamma_{DUT}))$$

Standard block diagram of an N-port VNA



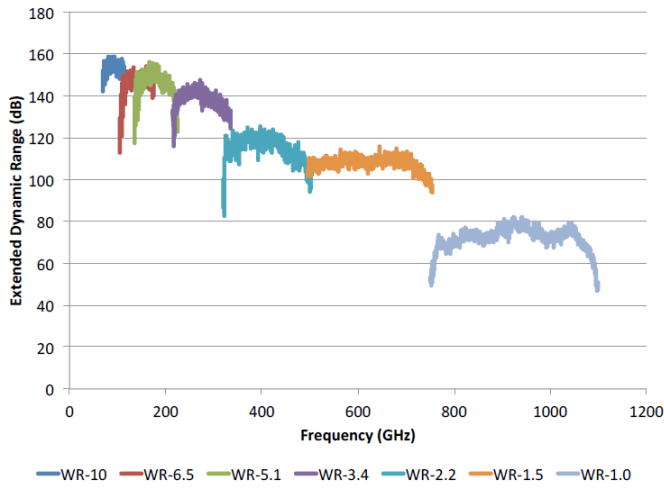
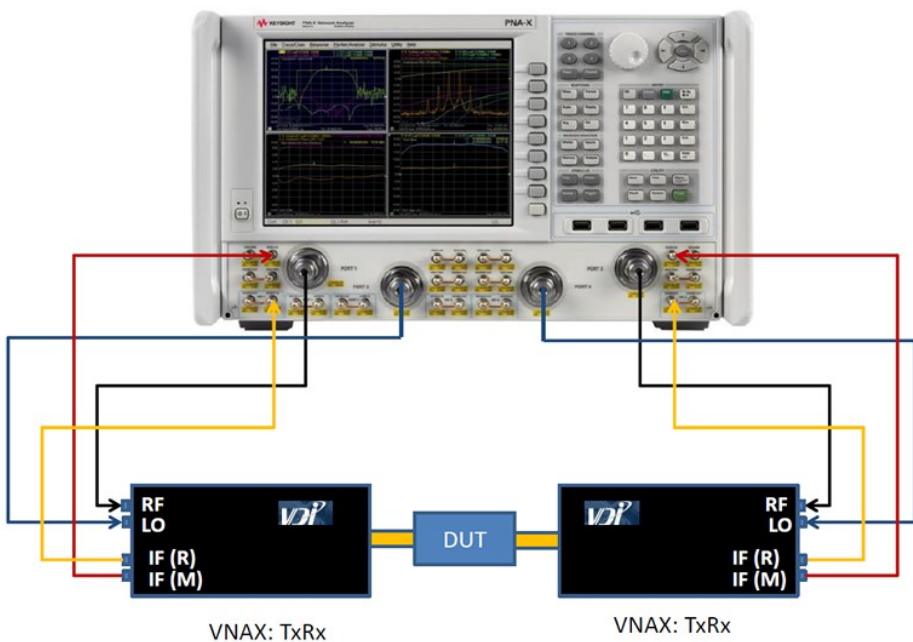
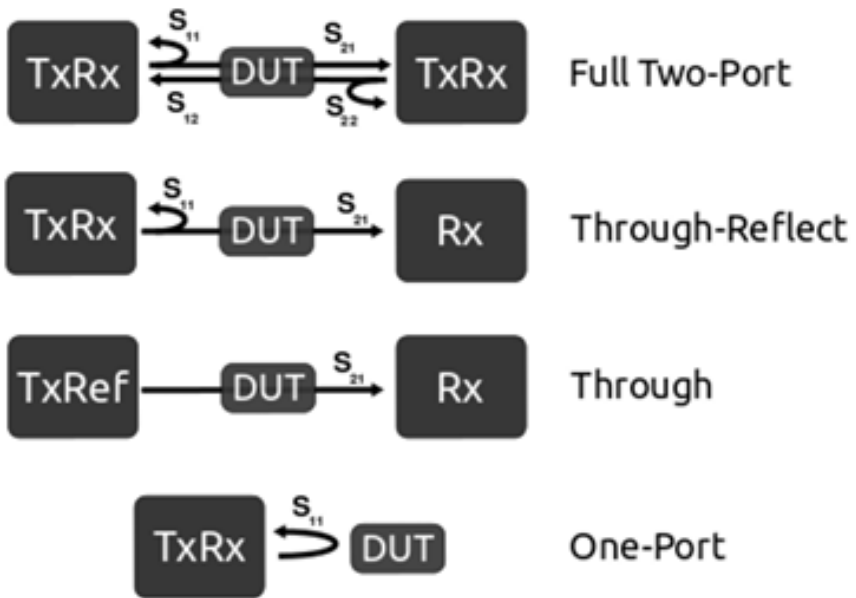
From Rohde & Schwarz

- Each test set separates incident and reflected wave at the test port
- Two sources: The RF and LO signal. The RF signal is switched between the test ports which is then operated as an active test port
- Each test set is combined with two separate receives for the measurement channel and the reference channel

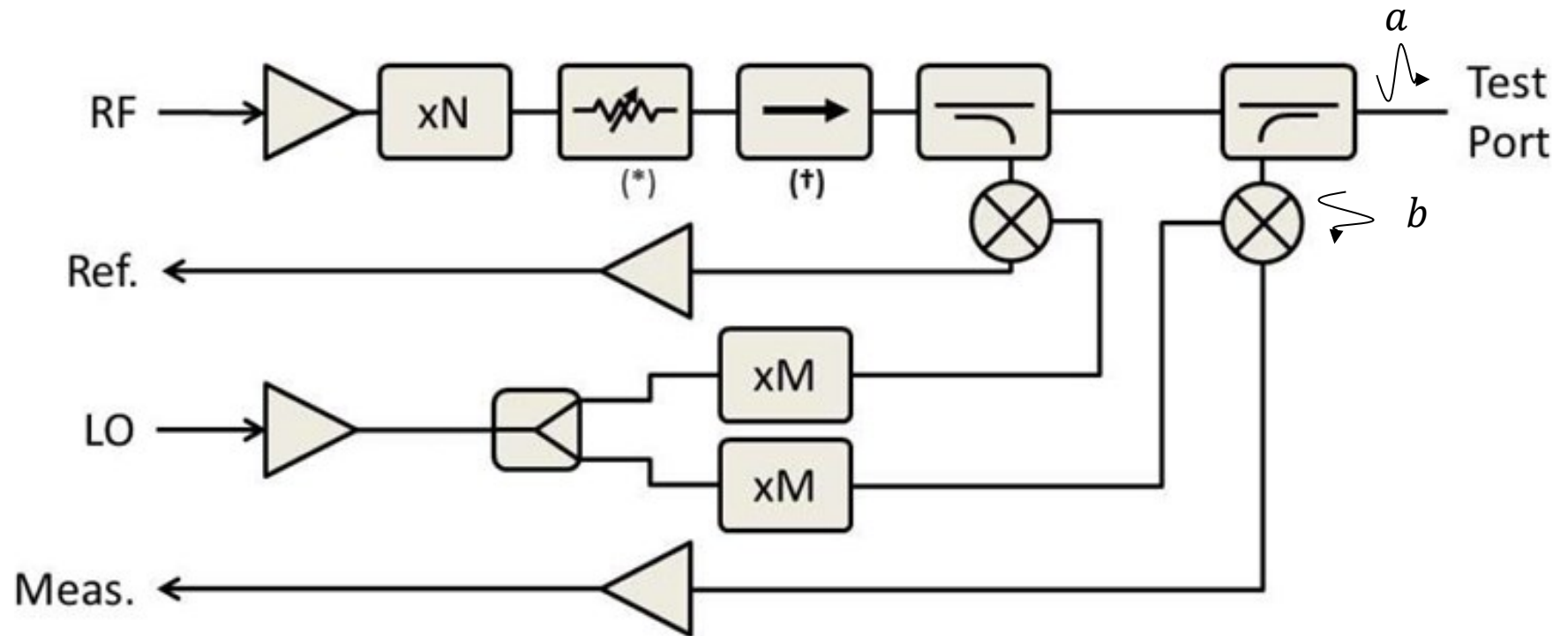
Frequency Extenders



Possible VNAX Configurations



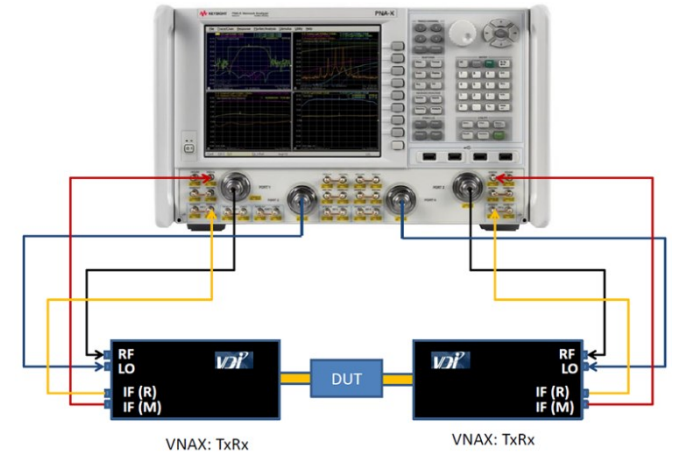
Frequency Extenders



Calibration

- Calibration kits provided by manufacturer in accordance with the Standards of Calibration
- One calibration kit for each connector type
- Repeated over regular intervals (i.e. temperature stability, quality of cables...)
- Modifying the channel settings can make calibration invalid (# points, start/end frequency...)

Calibration kit for 3.5mm connectors

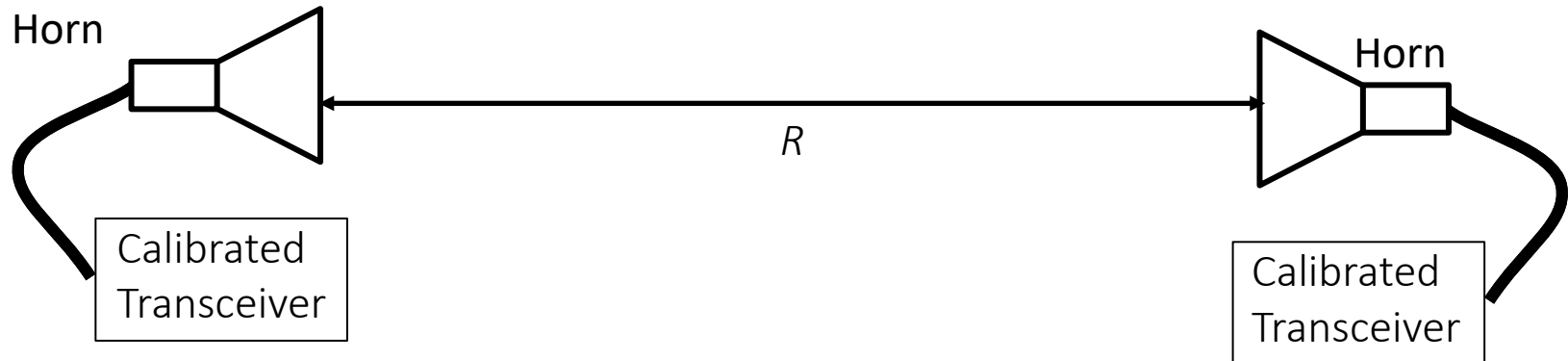


Calibration kit at WR5.1



SOLT calibration: Short – Open – Load – Through⁹

Measurement #1: Horn to Horn



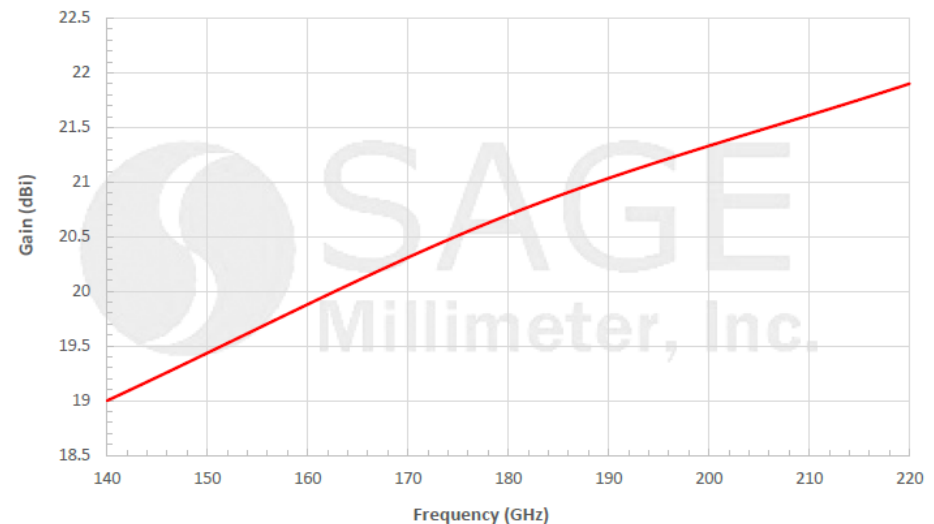
- Gain vs frequency.
- Apply time-gating techniques to reduce the multiple reflections of the environment

SAR-2013-05-S2

WR-05 Pyramidal Horn Antenna, 20 dBi Gain

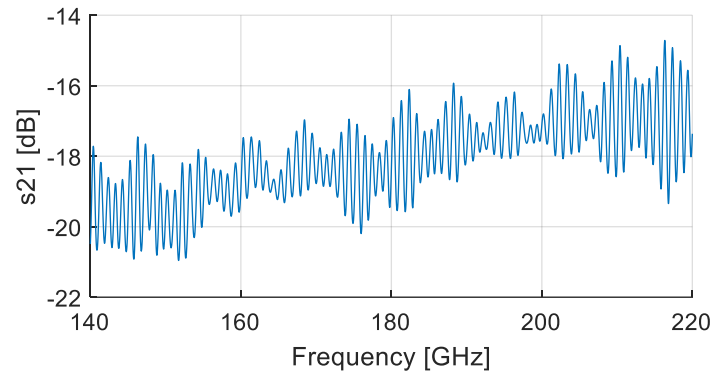


Typical Gain vs. Frequency



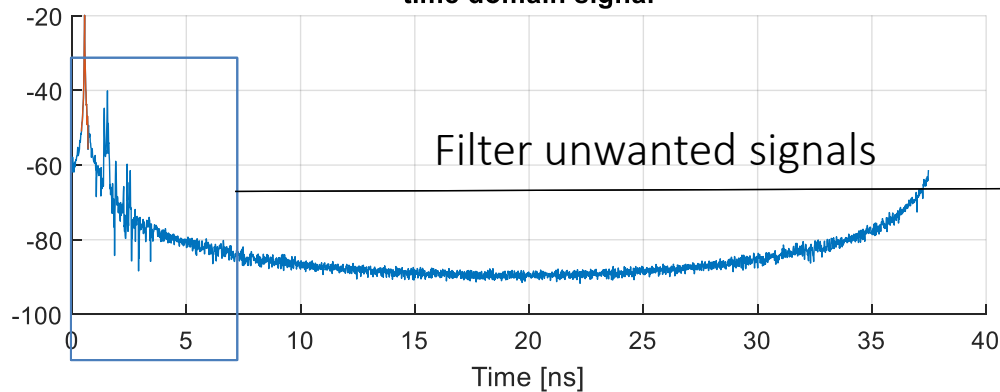
Time-gating

Measured Frequency Domain Spectrum

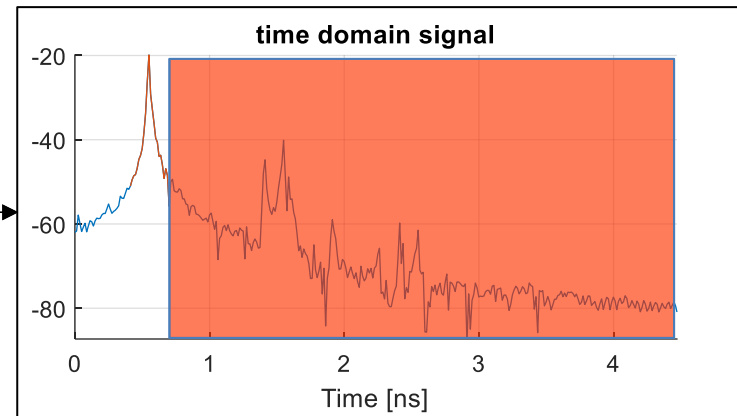


iFFT

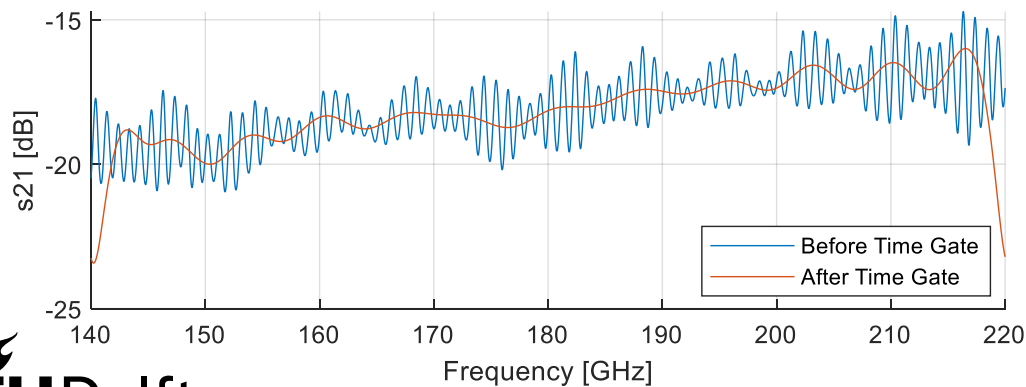
time domain signal



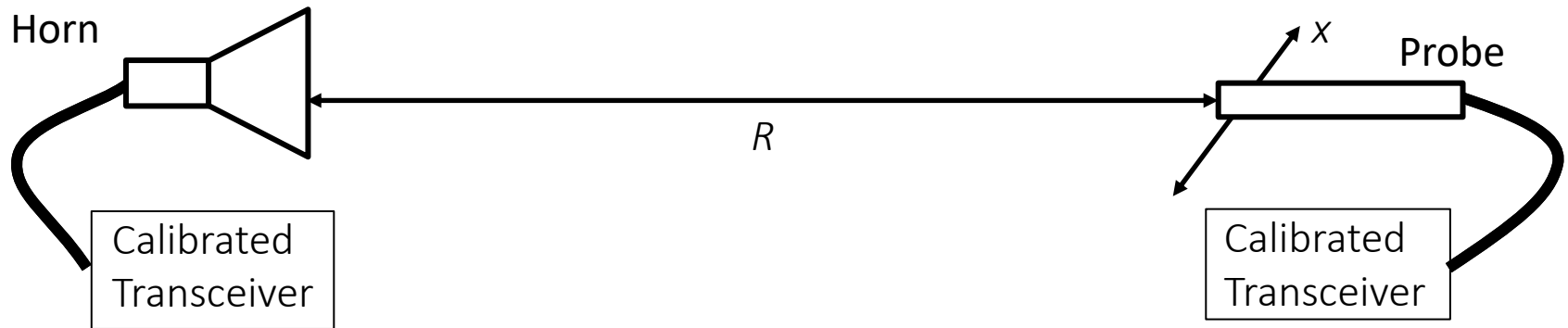
time domain signal



FFT

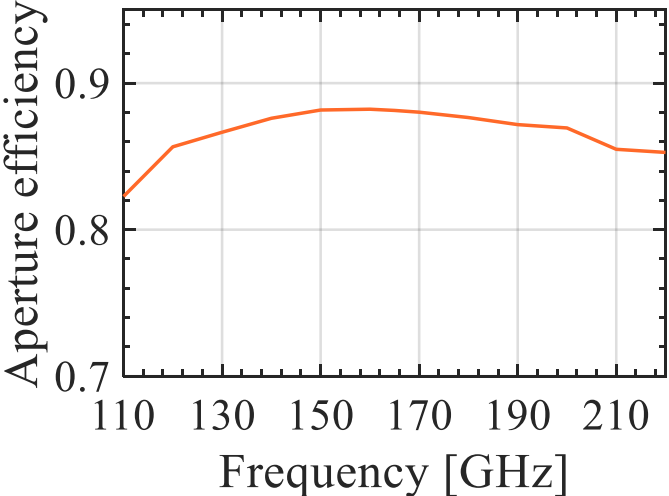
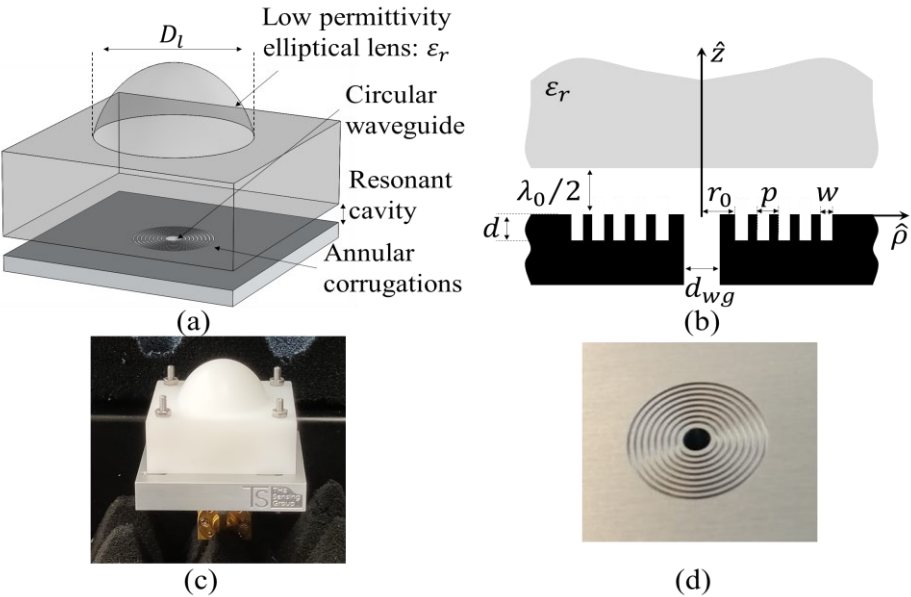


Measurements #2: Horn to Probe



- The radiation pattern of the horn. Comparison with airy pattern.
- Directivity vs frequency. Include directivity of ideal antenna (airy)
- Gain of the probe vs frequency. (Use the gain of the horn from Measurement 1)
- Apply time-gating techniques to reduce the multiple reflections of the environment

Lens Antenna



Center frequency $f_c = 165GHz$

Lens Diameter $D_L = 17.5mm$

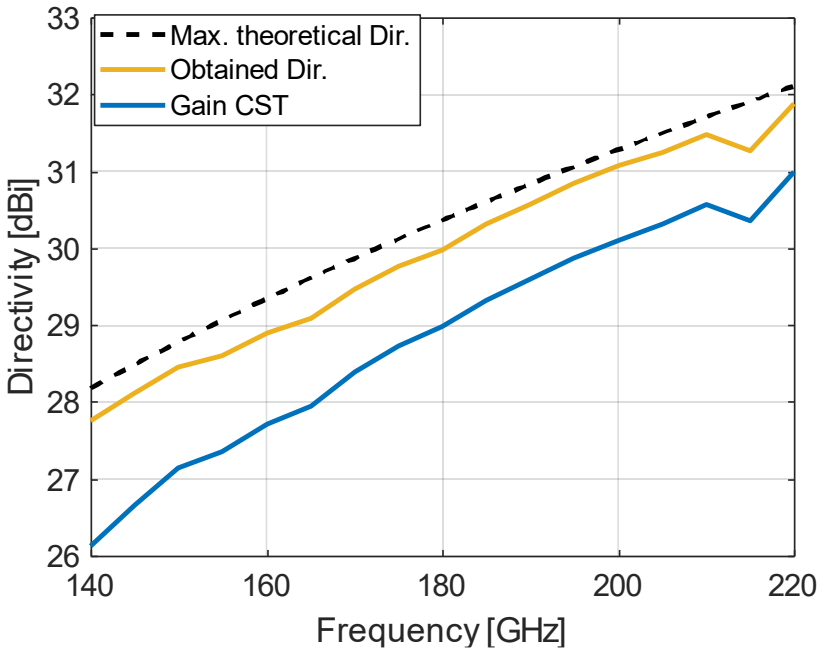
Lens subtended angle $\theta_{st} = 36.5^\circ$

Phase Centre $\Delta z = -1.2mm$

Elliptical parameters:

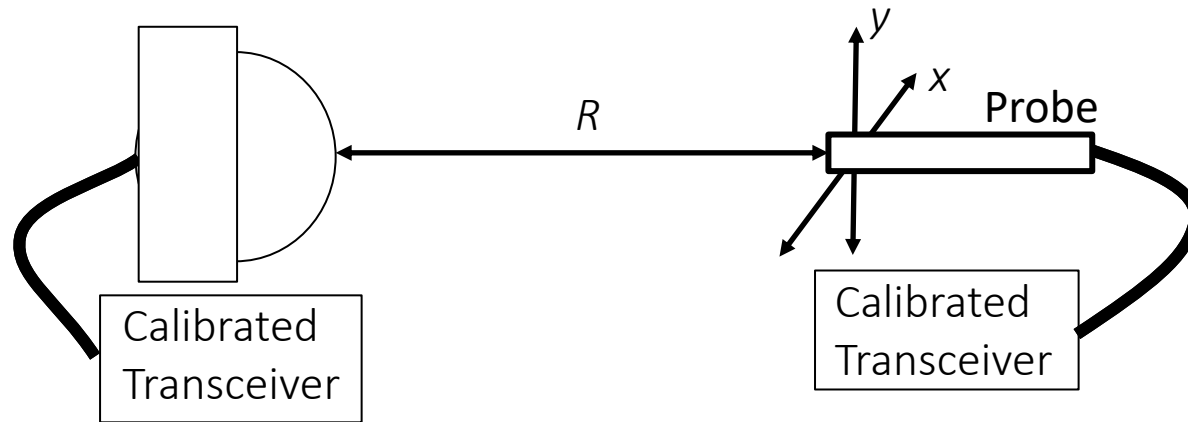
$a = 12.23mm$

$b = 9.20mm$



Gain includes the Ohmic loss from the metal block and plastic

Measurement #3: Lens to Probe

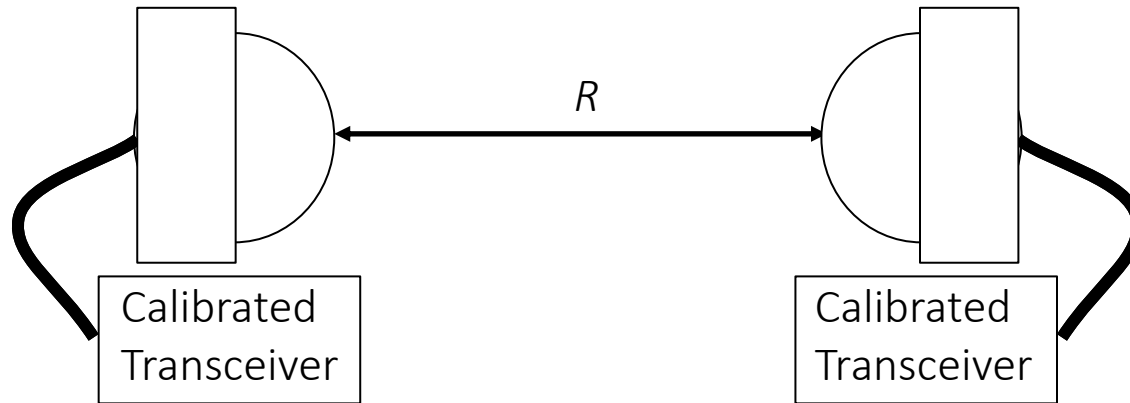


- The radiation pattern of the lens-antenna. Comparison with airy pattern.
- Directivity vs frequency. Include directivity of ideal antenna (airy)
- Apply time-gating techniques to reduce the multiple reflections of the environment

Test files in brightspace:

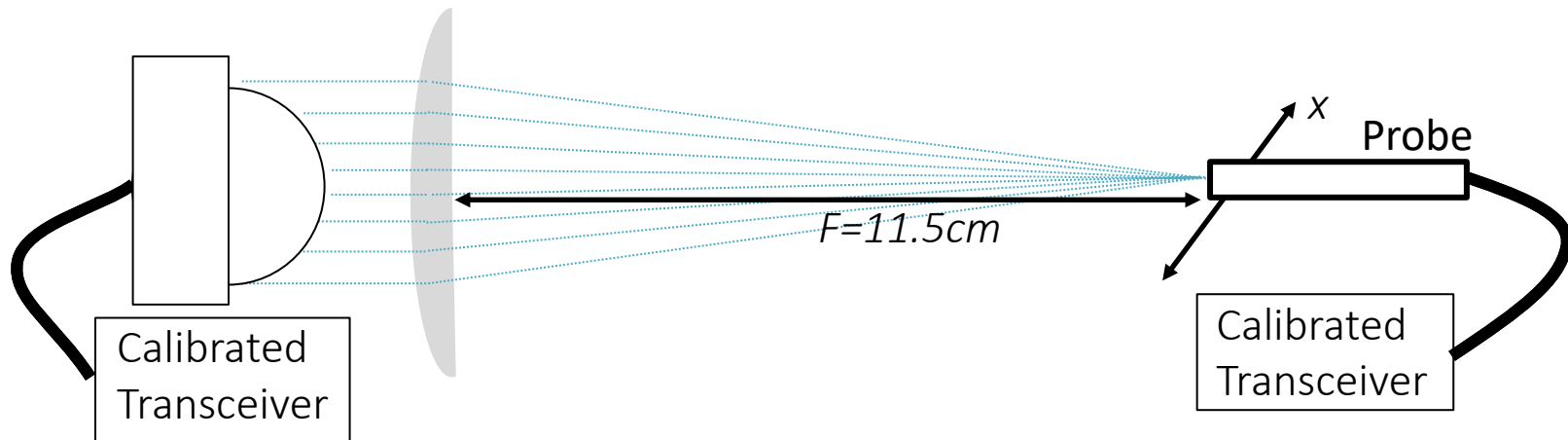
- .mat file of the measurement of a radiation pattern of the lens antenna

Measurement #4: Lens to Lens



- Coupling vs Frequency
- Apply time-gating techniques to reduce the multiple reflections of the environment

Measurement #5: Lens to Lens to Probe



- The radiation pattern of the horn. Comparison with airy pattern.
- Directivity vs frequency. Include directivity of ideal antenna (airy)
- Gain of the probe vs frequency. (Use the gain of the horn from Measurement 1)
- Apply time-gating techniques to reduce the multiple reflections of the environment